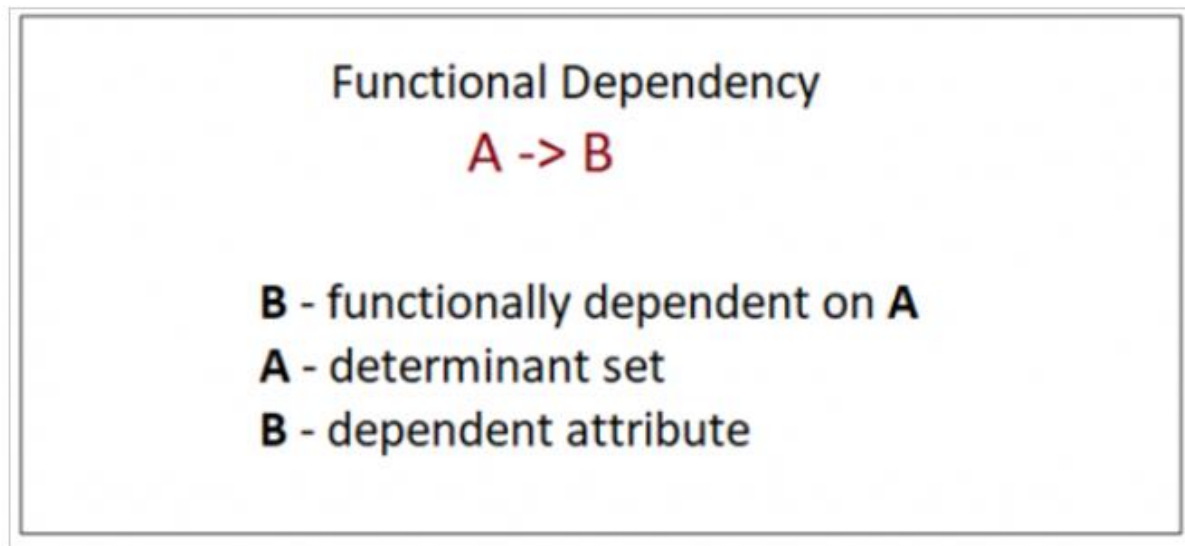


Functional Dependency

The functional dependency is a relationship that exists between two attributes. It typically exists between the primary key and non-key attribute within a table.



We have a <**Department**> table with two attributes – **Dept_Id** and **DeptName**.

DeptId = Department ID **DeptName** = Department Name

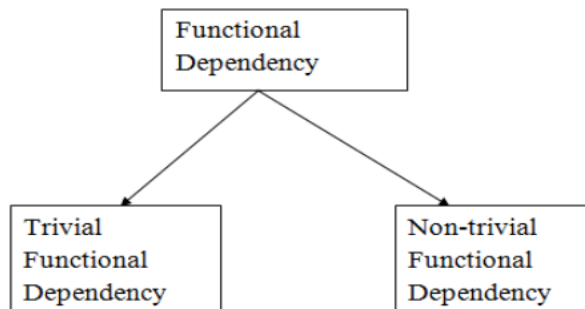
The **DeptId** is our primary key. Here, **DeptId** uniquely identifies the **DeptName** attribute. This is because if you want to know the department name, then at first you need to have the **DeptId**.

DeptId	DeptName
001	Finance
002	Marketing
003	HR

Therefore, the above functional dependency between **DeptId** and **DeptName** can be determined as **DeptId** is functionally dependent on **DeptName** –

DeptId -> DeptName

Types of Functional Dependency



1. Trivial functional dependency

- $A \rightarrow B$ has trivial functional dependency if B is a subset of A .
- The following dependencies are also trivial like: $A \rightarrow A$, $B \rightarrow B$

Example:

1. Consider a table with two columns Employee_Id and Employee_Name.
2. $\{Employee_id, Employee_Name\} \rightarrow Employee_Id$ is a trivial functional dependency as
3. Employee_Id is a subset of $\{Employee_Id, Employee_Name\}$.
4. Also, $Employee_Id \rightarrow Employee_Id$ and $Employee_Name \rightarrow Employee_Name$ are trivial dependencies

2. Non-trivial functional dependency

- $A \rightarrow B$ has a non-trivial functional dependency if B is not a subset of A .

- When A Intersection B is NULL, then $A \rightarrow B$ is called as complete non-trivial.

Example:

1. $ID \rightarrow Name$,
2. $Name \rightarrow DOB$

Inference Rule (IR):

- The Armstrong's axioms are the basic inference rule.
- Armstrong's axioms are used to conclude functional dependencies on a relational database.
- Using the inference rule, we can derive additional functional dependency from the initial set.

The Functional dependency has 6 types of inference rule:

1. Reflexive Rule (IR_1)

In the reflexive rule, if Y is a subset of X, then X determines Y.

1. If $X \supseteq Y$ then $X \rightarrow Y$

Self-determine $SID \rightarrow SID$ [Always valid]

Example:

$X = \{a, b, c, d, e\}$

2. Augmentation Rule (IR₂)

The augmentation is also called as a partial dependency. In augmentation, if X determines Y, then XZ determines YZ for any Z.

If $X \rightarrow Y$ then $XZ \rightarrow YZ$

Example:

$SID \rightarrow Sname$

$SID, Ph_No \rightarrow Sname, Pn_No$

1. For R (ABCD), if $A \rightarrow B$ then $AC \rightarrow BC$

3. Transitive Rule (IR₃)

In the transitive rule, if X determines Y and Y determine Z, then X must also determine Z.

$SID \rightarrow Sname$ and $Sname \rightarrow City$ then $SID \rightarrow city$

1. If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$

4. Union Rule (IR₄)

Union rule says, if X determines Y and X determines Z, then X must also determine Y and Z.

1. If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow YZ$

5. Decomposition Rule (IR₅)

Decomposition rule is also known as project rule. It is the reverse of union rule.

This Rule says, if X determines Y and Z, then X determines Y and X determines Z separately.

If $X \rightarrow YZ$ then $X \rightarrow Y$ and $X \rightarrow Z$

6. Pseudo transitive Rule (IR₆)

In Pseudo Transitive Rule, if X determines Y and YZ determines W, then XZ determines W.

1. If $X \rightarrow Y$ and $WY \rightarrow Z$ then $XW \rightarrow Z$

7. Composition

If $X \rightarrow Y$ and $Z \rightarrow W$

Then $XZ \rightarrow YW$